

Supplementary Derivation for the Paper Fault-Aware Design for Reconfigurable Holographic Surface-Aided ISAC Systems

Lu Wang*, Mohamadreza Delbari*, Gui Zhou[†], Luis F. Abanto-Leon[‡],

Matthias Hollick*, and Vahid Jamali*

*Technical University of Darmstadt, Darmstadt, Germany,

[†]Huazhong University of Science and Technology, Wuhan, China,

[‡]Ruhr University Bochum, Bochum, Germany

This supplementary document provides the detailed derivations of $\mathbf{A}_{\eta_0}$ and $\mathbf{B}_{\eta_0}$ used for the paper *L. Wang, M. Delbari, G. Zhou, L. F. Abanto-Leon, M. Hollick, and V. Jamali, "Fault-Aware Design for Reconfigurable Holographic Surface-Aided ISAC Systems," 2026 IEEE International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC), 2026*. These derivations are omitted from the main paper due to space limitations.

Please cite the following paper if this document or its derivations are used:

```
@INPROCEEDINGS{bLuWangRHS,
  author={Lu Wang and Delbari, Mohamadreza and Zhou, Gui and Abanto-Leon, Luis F. and
    Hollick, Matthias and Jamali, Vahid},
  booktitle={2026 IEEE International Symposium on Personal, Indoor and Mobile Radio
    Communications},
  title={Fault-Aware Design for Reconfigurable Holographic Surface-Aided ISAC Systems},
  year={2026},
  volume={},
  number={},
  pages={},
}
```

Note that the equation numbering in this document follows the numbering sequence used in the main paper. Detailed derivations are shown below.

The calculation of MCRB is dependent on the derivation of $\mathbf{A}_{\eta}$ and $\mathbf{B}_{\eta}$ given in (14) and (15) in the main paper, in which the first-order and second-order partial derivatives of $\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})$ with respect to $\boldsymbol{\eta}$ are needed. To achieve this goal, $\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta}) = \text{vec}(\mathbf{G}\tilde{\boldsymbol{\Psi}}\mathbf{Q}\mathbf{X})$ is further written as follows.

$$\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta}) = \alpha \text{vec} \left(\mathbf{a}(\phi_e, \phi_a) \mathbf{a}^H(\phi_e, \phi_a) \tilde{\boldsymbol{\Psi}} \mathbf{Q} \mathbf{X} \right). \quad (32)$$

Denoting $\mathbf{\Omega} = \mathbf{a}(\phi_e, \phi_a) \mathbf{a}^H(\phi_e, \phi_a)$ and $\boldsymbol{\eta} = [\boldsymbol{\phi}^T, \tilde{\boldsymbol{\alpha}}^T]^T$ where $\boldsymbol{\phi} = [\phi_e, \phi_a]^T$ and $\tilde{\boldsymbol{\alpha}} = [\text{Re}(\alpha), \text{Im}(\alpha)]^T$, the first-order partial derivative is obtained as follows

$$\frac{\partial \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi}} = \begin{bmatrix} \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \alpha \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix}, \quad (33)$$

$$\frac{\partial \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}}} = \text{vec} \left(\mathbf{\Omega} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \begin{bmatrix} 1 \\ j \end{bmatrix}, \quad (34)$$

where $\dot{\mathbf{\Omega}}_{\phi_e}$ and $\dot{\mathbf{\Omega}}_{\phi_a}$ are given as follows

$$\dot{\mathbf{\Omega}}_{\phi_e} = \frac{\partial \mathbf{\Omega}}{\partial \phi_e} = j\kappa \left[d_y \sin(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) - d_z \cos(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Z) \right) \right], \quad (35)$$

$$\dot{\mathbf{\Omega}}_{\phi_a} = \frac{\partial \mathbf{\Omega}}{\partial \phi_a} = -j\kappa d_y \cos(\phi_e) \cos(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right), \quad (36)$$

$\mathbf{a}$ stands for $\mathbf{a}(\phi_e, \phi_a)$ for the sake of notational convenience. $\mathbf{r}_Y = [0, 1, \dots, R_y - 1]^T$ and $\mathbf{r}_Z = [0, 1, \dots, R_z - 1]^T$ collect the index of y-axis and z-axis, respectively. Next, we take the second-order partial derivative of $\tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})$ with respect to $\boldsymbol{\eta}$, which can be further partitioned as

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\eta}^2} = \begin{bmatrix} \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi}^2} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}} \partial \boldsymbol{\phi}} \\ \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi} \partial \tilde{\boldsymbol{\alpha}}} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}}^2} \end{bmatrix}, \quad (37)$$

with three elements in (37) given as follows

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi} \partial \tilde{\boldsymbol{\alpha}}} = \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}} \partial \boldsymbol{\phi}} = \begin{bmatrix} 1 \\ j \end{bmatrix} \begin{bmatrix} \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \text{vec} \left(\dot{\mathbf{\Omega}}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix}^T, \quad (38)$$

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \tilde{\boldsymbol{\alpha}}^2} = \mathbf{0}_2, \quad (39)$$

$$\frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \boldsymbol{\phi}^2} = \begin{bmatrix} \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_e^2} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_a \partial \phi_e} \\ \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_e \partial \phi_a} & \frac{\partial^2 \tilde{\boldsymbol{\mu}}(\boldsymbol{\eta})}{\partial \phi_a^2} \end{bmatrix} = \begin{bmatrix} \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_e \phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_a \phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_e \phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \alpha \text{vec} \left(\ddot{\mathbf{\Omega}}_{\phi_a \phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix}, \quad (40)$$

where $\ddot{\mathbf{\Omega}}_{\phi_e \phi_e}$, $\ddot{\mathbf{\Omega}}_{\phi_e \phi_a}$, $\ddot{\mathbf{\Omega}}_{\phi_a \phi_e}$, and $\ddot{\mathbf{\Omega}}_{\phi_a \phi_a}$ are derived as follows

$$\begin{aligned} \ddot{\mathbf{\Omega}}_{\phi_e \phi_e} &= j\kappa \left[d_y \cos(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) + d_z \sin(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Z) \right) \right. \\ &\quad \left. + d_y \sin(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \dot{\mathbf{\Omega}}_{\phi_e} - \dot{\mathbf{\Omega}}_{\phi_e} \text{diag}(\mathbf{r}_Y) \right) - d_z \cos(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \dot{\mathbf{\Omega}}_{\phi_e} - \dot{\mathbf{\Omega}}_{\phi_e} \text{diag}(\mathbf{r}_Z) \right) \right], \end{aligned} \quad (41)$$

$$\begin{aligned} \ddot{\mathbf{\Omega}}_{\phi_e \phi_a} &= \ddot{\mathbf{\Omega}}_{\phi_a \phi_e} = j\kappa \left[d_y \sin(\phi_e) \cos(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) \right. \\ &\quad \left. + d_y \sin(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \dot{\mathbf{\Omega}}_{\phi_a} - \dot{\mathbf{\Omega}}_{\phi_a} \text{diag}(\mathbf{r}_Y) \right) - d_z \cos(\phi_e) \left(\text{diag}(\mathbf{r}_Z) \dot{\mathbf{\Omega}}_{\phi_a} - \dot{\mathbf{\Omega}}_{\phi_a} \text{diag}(\mathbf{r}_Z) \right) \right], \end{aligned} \quad (42)$$

$$\ddot{\mathbf{\Omega}}_{\phi_a \phi_a} = j\kappa \left[d_y \cos(\phi_e) \sin(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \mathbf{a} \mathbf{a}^H - \mathbf{a} \mathbf{a}^H \text{diag}(\mathbf{r}_Y) \right) - d_y \cos(\phi_e) \cos(\phi_a) \left(\text{diag}(\mathbf{r}_Y) \dot{\mathbf{\Omega}}_{\phi_a} - \dot{\mathbf{\Omega}}_{\phi_a} \text{diag}(\mathbf{r}_Y) \right) \right]. \quad (43)$$

Then, we partition $\mathbf{A}_\eta$ and $\mathbf{B}_\eta$ into 4 blocks, namely

$$\mathbf{A}_\eta = \begin{bmatrix} \mathbf{A}_{\phi\phi} & \mathbf{A}_{\alpha\phi} \\ \mathbf{A}_{\phi\alpha} & \mathbf{A}_{\alpha\alpha} \end{bmatrix}, \mathbf{B}_\eta = \begin{bmatrix} \mathbf{B}_{\phi\phi} & \mathbf{B}_{\alpha\phi} \\ \mathbf{B}_{\phi\alpha} & \mathbf{B}_{\alpha\alpha} \end{bmatrix}. \quad (44)$$

According to (14) and (15) in the main paper, each sub-block matrix in $\mathbf{B}_\eta$ is obtained as (45), (46), (47), and (48). Each sub-block matrix in $\mathbf{A}_\eta$ is obtained as (49), (50), (51), and (52).

$$\mathbf{B}_{\phi\phi} = \frac{2|\alpha|^2}{\sigma_s^2} \text{Re} \begin{bmatrix} \text{tr} \left(\dot{\Omega}_{\phi_e} \mathbf{F} \dot{\Omega}_{\phi_e}^H \right) & \text{tr} \left(\dot{\Omega}_{\phi_e} \mathbf{F} \dot{\Omega}_{\phi_a}^H \right) \\ \text{tr} \left(\dot{\Omega}_{\phi_a} \mathbf{F} \dot{\Omega}_{\phi_e}^H \right) & \text{tr} \left(\dot{\Omega}_{\phi_a} \mathbf{F} \dot{\Omega}_{\phi_a}^H \right) \end{bmatrix}, \quad (45)$$

$$\mathbf{B}_{\phi\alpha} = \frac{2}{\sigma_s^2} \text{Re} \begin{bmatrix} \alpha^H \text{tr} \left(\Omega \mathbf{F} \dot{\Omega}_{\phi_e}^H \right) & \alpha^H \text{tr} \left(\Omega \mathbf{F} \dot{\Omega}_{\phi_a}^H \right) \\ j \alpha^H \text{tr} \left(\Omega \mathbf{F} \dot{\Omega}_{\phi_e}^H \right) & j \alpha^H \text{tr} \left(\Omega \mathbf{F} \dot{\Omega}_{\phi_a}^H \right) \end{bmatrix}, \quad (46)$$

$$\mathbf{B}_{\alpha\phi} = \mathbf{B}_{\phi\alpha}^T, \quad (47)$$

$$\mathbf{B}_{\alpha\alpha} = \frac{2}{\sigma_s^2} \mathbf{I}_2 \text{tr} \left(\bar{\Omega} \mathbf{R}_x \bar{\Omega}^H \right), \quad (48)$$

$$\begin{aligned} \mathbf{A}_{\phi\phi} &= \frac{2}{\sigma_s^2} \text{Re} \begin{bmatrix} \varepsilon(\eta)^H \alpha \text{vec} \left(\ddot{\Omega}_{\phi_e\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\eta)^H \alpha \text{vec} \left(\ddot{\Omega}_{\phi_a\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \varepsilon(\eta)^H \alpha \text{vec} \left(\ddot{\Omega}_{\phi_e\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\eta)^H \alpha \text{vec} \left(\ddot{\Omega}_{\phi_a\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix} - \mathbf{B}_{\phi\phi} \\ &= \frac{2|\alpha|^2}{\sigma_s^2} \text{Re} \begin{bmatrix} \text{tr} \left(\ddot{\Omega}_{\phi_e\phi_e} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\ddot{\Omega}_{\phi_e\phi_e} \mathbf{F} \Omega^H \right) & \text{tr} \left(\ddot{\Omega}_{\phi_a\phi_e} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\ddot{\Omega}_{\phi_a\phi_e} \mathbf{F} \Omega^H \right) \\ \text{tr} \left(\ddot{\Omega}_{\phi_e\phi_a} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\ddot{\Omega}_{\phi_e\phi_a} \mathbf{F} \Omega^H \right) & \text{tr} \left(\ddot{\Omega}_{\phi_a\phi_a} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\ddot{\Omega}_{\phi_a\phi_a} \mathbf{F} \Omega^H \right) \end{bmatrix} - \mathbf{B}_{\phi\phi}. \end{aligned} \quad (49)$$

$$\begin{aligned} \mathbf{A}_{\phi\alpha} &= \frac{2}{\sigma_s^2} \text{Re} \begin{bmatrix} \varepsilon(\eta)^H \alpha \text{vec} \left(\dot{\Omega}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\eta)^H \alpha \text{vec} \left(\dot{\Omega}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \\ \varepsilon(\eta)^H j \alpha \text{vec} \left(\dot{\Omega}_{\phi_e} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) & \varepsilon(\eta)^H j \alpha \text{vec} \left(\dot{\Omega}_{\phi_a} \tilde{\Psi} \mathbf{Q} \mathbf{X} \right) \end{bmatrix} - \mathbf{B}_{\phi\alpha} \\ &= \frac{2|\alpha|^2}{\sigma_s^2} \text{Re} \begin{bmatrix} \text{tr} \left(\dot{\Omega}_{\phi_e} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\dot{\Omega}_{\phi_e} \mathbf{F} \Omega^H \right) & \text{tr} \left(\dot{\Omega}_{\phi_a} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\dot{\Omega}_{\phi_a} \mathbf{F} \Omega^H \right) \\ j \left(\text{tr} \left(\dot{\Omega}_{\phi_e} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\dot{\Omega}_{\phi_e} \mathbf{F} \Omega^H \right) \right) & j \left(\text{tr} \left(\dot{\Omega}_{\phi_a} \bar{\mathbf{F}} \Omega^H \right) - \text{tr} \left(\dot{\Omega}_{\phi_a} \mathbf{F} \Omega^H \right) \right) \end{bmatrix} - \mathbf{B}_{\phi\alpha}. \end{aligned} \quad (50)$$

$$\mathbf{A}_{\alpha\phi} = \mathbf{A}_{\phi\alpha}^T, \quad (51)$$

$$\mathbf{A}_{\alpha\alpha} = -\mathbf{B}_{\alpha\alpha}, \quad (52)$$

where

$$\mathbf{F} = \tilde{\Psi} \mathbf{Q} (\mathbf{W} \mathbf{W}^H + \mathbf{R}_s) \mathbf{Q}^H \tilde{\Psi}^H. \quad (53)$$

$$\bar{\mathbf{F}} = \tilde{\Psi} \mathbf{Q} (\mathbf{W} \mathbf{W}^H + \mathbf{R}_s) \mathbf{Q}^H \tilde{\Psi}^H. \quad (54)$$

With all the above derivations, we can obtain both $\mathbf{A}_{\eta_0}$ and $\mathbf{B}_{\eta_0}$ by letting $\eta = \eta_0$.